

# AI Code Cleaner: Feature Report & Current Status

This document provides a detailed overview of the core features implemented in the **AI Code Cleaner Tool** and the current state of the project based on the latest automated test runs.

## Core Features

The tool now includes six powerful modular analysis services:

### 1. Semantic Clone Detection

- **What it does:** Identified duplicate or near-duplicate logic across the project.
- **AI-Powered:** Uses AST parsing and semantic hashing to catch clones that are logically identical but textually different.
- **Status:** Fully Functional. Detected **12 clone groups** in the latest run.

### 2. Unused Function Detection

- **What it does:** Scans the codebase to find methods that are defined but never called.
- **Status:** Fully Functional. Identified **2 unused functions** in the latest run.

### 3. Comment & Javadoc Remover

- **What it does:** Detects and optionally clears redundant comments, block comments, and Javadocs.
- **Detail:** Distinguishes between standard comments and `TODO/FIXME` tags.
- **Status:** Fully Functional. Detected **36 comments** (including 3 `TODO`s).

### 4. Whitespace & Formatting Cleaner

- **What it does:** Finds trailing spaces and consecutive blank lines.
- **Status:** Fully Functional. Detected **4 issues**.

### 5. Anti-Pattern Detector

- **What it does:** Scans for common Java “code smells” and bad practices.
- **Metrics:**
  - **Magic Numbers:** 63 detected
  - **System.out.println:** 8 detected
  - **Empty Catch Blocks:** 1 detected
  - **God Methods:** 1 detected (Complex functions)

- **String == Checks:** 1 detected
- **Status:** Fully Functional. Identified **76 total anti-patterns**.

## 6. Unused & Wildcard Import Detector

- **What it does:** identifies **unused** imports and discouraged **\*** (wildcard) imports.
  - **Status:** Fully Functional. Detected **5 import issues**.
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## Current Progress “How We Doing”

- **Architecture:** The tool is now highly modular. Each feature resides in its own service (e.g., `AntiPatternDetector.java`, `UnusedImportDetector.java`).
- **Stability:** Recent encoding and `pom.xml` issues have been resolved. The tool builds successfully using Maven or the provided `.bat` script.
- **UI:** The modern dark-mode UI (FlatLaf) is fully integrated with a dashboard to visualize these metrics.
- **Safety:** The `RefactorService` handles deletions safely, preserving character offsets and cleaning up surrounding blank lines.